Failure Analysis: Technical Worksheet

Grade 10 · Worksheet · About 25 minutes
Work with pen and paper. Use color pencils if you like. You do not need a computer or the internet.

Name: _____

Grade/Section: _____

Date: _____

> **How to use this worksheet**
> - Do the parts **in order** from top to bottom.
> - Each part has a **time** so you know how long to spend.
> - Read every instruction before you write. You can do this on your own.

Part 1 – What you will learn (about 3 minutes)

What you will learn: Failure analysis is a structured way to understand why a system stopped working as expected. You separate **symptoms** from **causes**, isolate variables, use error clues, and document findings so the same failure is less likely to repeat.

Do this:

1. **Describe one failure** you have seen (app crash, Wi-Fi drop, printer jam, login error).

   _____

2. In a few words, what was the **symptom** people noticed first?

   _____

Part 2 – Symptoms vs causes (about 5 minutes)

A **symptom** is what users observe (slow, blank screen, error code). A **cause** is the underlying condition (full disk, faulty cable, buggy update, misconfiguration). Treating only symptoms (rebooting forever) without finding causes invites repeats.

> **Real-life scenario:** Students say “the projector is broken.” The symptom is no image. Possible causes: sleep mode, loose HDMI, wrong input source, dead bulb, or laptop output disabled.

Discuss with a partner or reflect in writing: Why is “the projector is broken” too vague for analysis?

Do this: Mark each statement **S** (symptom) or **C** (possible cause).

Statement	S or C?
The login page never loads	
DNS settings point to an unreachable server	
“Disk almost full” warning ignored for weeks	
Files take minutes to open	

Part 3 – Isolate and gather clues (about 5 minutes)

Good analysis **isolates** factors: Does it happen on one device or many? One account or all? Wired and wireless? After a change (update, new cable)? Capture **error messages**, times, and recent changes. Guessing without evidence wastes effort.

> **Real-life scenario:** Only Room 12 cannot print. Rooms 11 and 13 work. The shared printer is online. Someone swapped Room 12's network cable yesterday during a desk move.

Critical thinking: Which clue most strongly points you toward the cable or local connection first? Why?

Do this: List three questions you would ask to isolate a “Wi-Fi keeps dropping” report.

1. _____
2. _____
3. _____

Part 4 – Hardware, software, and documentation (about 5 minutes)

Failures may be **hardware** (physical parts), **software** (apps/OS), **configuration**, or **human process** (wrong steps). Writing a short **incident note**—what happened, evidence, likely cause, fix, prevention—turns a one-off repair into organizational learning.

> **Real-life scenario:** A club laptop overheats and shuts down. Dust blocks the vents (hardware/environment). Fans spin at 100%. After cleaning, the team also writes: "Monthly vent check; don't block side vents with binders."

Do this: Classify each failure mainly as **Hardware**, **Software**, or **Process** (best fit).

Failure	Category
App crashes after a rushed update with no test	_____
Frayed power cable arcs and cuts power	_____
Two people overwrite the same file because there is no naming rule	_____

Write a 3-line incident note for one failure from Part 1 (symptom / likely cause / prevention).

Part 5 – Analysis flow sketch (about 5 minutes)

Do this:

1. **Draw a flow**: Observe symptom → Gather clues → Form hypothesis → Test/isolate → Fix → Document prevention.



2. **Star the step** teams most often skip, and say why skipping it is costly.

Part 6 – Check what you learned (about 2 minutes)

Checklist. Put a ✓ in the box when it is true for you.

☐ I can separate **symptoms** from likely **causes**.

☐ I can ask isolating questions and use error/time/change clues.

☐ I can classify failures as hardware, software, or process-related.

☐ I can write a short **incident note** with prevention.

☐ I can sketch an analysis flow from observe to document.

When all boxes are checked, you have reviewed the main ideas from this worksheet.

Worksheet on Failure Analysis (Technical). Central topic: Failure analysis. Main technology area: Troubleshooting and system reliability.